

ConfigMaps&Secrets

Starting MiniKube & Creating a Configmap:

```
hp@Dharanidhar MINGW64 ~
$ cd /C/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4

hp@Dharanidhar MINGW64 /C/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ minikube start
* minikube v1.38.1 on Microsoft Windows 11 Pro 25H2
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Restarting existing docker container for "minikube" ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default

hp@Dharanidhar MINGW64 /C/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

hp@Dharanidhar MINGW64 /C/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ vim configmap.yaml

hp@Dharanidhar MINGW64 /C/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ cat configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: project-config
data:
  App_Name: "COTY AMS"
  App_Mode: "Development"

hp@Dharanidhar MINGW64 /C/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ kubectl apply -f configmap.yaml
configmap/project-config created
```

Creating a Secret using stringdata to avoid manual base64 encoding:

```
hp@Dharanidhar MINGW64 /C/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ vim secret.yaml

hp@Dharanidhar MINGW64 /C/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ cat secret.yaml
apiVersion: v1
kind: Secret
metadata:
  name: project-secret
type: Opaque
stringData:
  App_Password: "Cotyjuly@2026"

hp@Dharanidhar MINGW64 /C/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ kubectl apply -f secret.yaml
secret/project-secret created
```

Creating a Pod using configmaps and Secret:

```
hp@Dharanidhar MINGW64 /C:/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ vim pod.yaml

hp@Dharanidhar MINGW64 /C:/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ cat pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: project-config-secrete
spec:
  containers:
  - name: nginx
    image: nginx:1.27
    env:
    - name: App_Name
      valueFrom:
        configMapKeyRef:
          name: project-config
          key: App_Name
    - name: App_Mode
      valueFrom:
        configMapKeyRef:
          name: project-config
          key: App_Mode
    - name: App_Password
      valueFrom:
        secretKeyRef:
          name: project-secret
          key: App_Password

hp@Dharanidhar MINGW64 /C:/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ kubectl apply -f pod.yaml
pod/project-config-secrete created
```

Verifying all the resources using commands:

```
hp@Dharanidhar MINGW64 /C:/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ kubectl get configmaps
NAME          DATA   AGE
kube-root-ca.crt  1      22d
project-config  2      33m

hp@Dharanidhar MINGW64 /C:/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ kubectl get secrets
NAME          TYPE      DATA   AGE
project-secret Opaque    1       20m

hp@Dharanidhar MINGW64 /C:/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
project-config-secrete 1/1     Running   0          3m54s
```

```
hp@Dharanidhar MINGW64 /C:/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ kubectl describe configmap project-config
Name:         project-config
Namespace:    default
Labels:       <none>
Annotations:  <none>

Data
====
App_Mode:
----
Development

App_Name:
----
COTY AMS

BinaryData
====

Events: <none>

hp@Dharanidhar MINGW64 /C:/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ kubectl describe secret project-secret
Name:         project-secret
Namespace:    default
Labels:       <none>
Annotations:  <none>

Type: Opaque

Data
====
App_Password: 13 bytes

hp@Dharanidhar MINGW64 /C:/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ ^C

hp@Dharanidhar MINGW64 /C:/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ kubectl get secret project-secret -o jsonpath='{.data.App_Password}' | base64 --decode
Cotyjuly@2026
hp@Dharanidhar MINGW64 /C:/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
```

```

hp@Dharanidhar MINGW64 /C:/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ kubectl get all
NAME                                READY    STATUS    RESTARTS   AGE
pod/project-config-screte          1/1      Running   0           9m19s

NAME                                TYPE          CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
service/kubernetes                 ClusterIP      10.96.0.1     <none>        443/TCP    22d

hp@Dharanidhar MINGW64 /C:/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ kubectl delete all
error: resource(s) were provided, but no name was specified

hp@Dharanidhar MINGW64 /C:/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ kubectl delete configmap project-config
configmap "project-config" deleted from default namespace

hp@Dharanidhar MINGW64 /C:/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ kubectl delete secret project-secret
secret "project-secret" deleted from default namespace

hp@Dharanidhar MINGW64 /C:/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ kubectl delete -f pod.yaml
pod "project-config-screte" deleted from default namespace

hp@Dharanidhar MINGW64 /C:/Users/hp/gitrepository/te-5th_and_6th_batch_assignments/Pavithra/K8S-Assignment-4 (main)
$ kubectl get all
NAME                                TYPE          CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
service/kubernetes                 ClusterIP      10.96.0.1     <none>        443/TCP    22d

```

NAME:service/Kubernetes is an built in Kubernetes API-Service.